Markups and Public Procurement: Evidence from Czech Construction Tenders

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Abstract

This paper analyzes the effect of public procurement on firm markups, using a panel dataset of Czech construction firms from 2006 to 2021. Markups are estimated through a structural framework, while biases are addressed using a selection on observables design and models for unobserved factors. Propensity score-based estimations indicate that firm markups increase by approximately 15% during contract years. A temporal analysis, employing synthetic control and matrix completion methods, reveals that treatment effects decline from around 30% in 2006 to 10% in 2021. These patterns are consistent with institutional improvements in the Czech Republic and offer empirical evidence of increasing efficiency in public spending.

Keywords: Models with Panel Data; Firm Behavior: Empirical Analysis; Procurement; Construction

JEL: C23, D22, H57, L74

[1]Predoctoral Fellow, Tobin Center for Economic Policy, Yale University (marek.chadim@yale.edu; https://marek-chadim.github.io). This paper is abridged and revised from my MSc thesis, submitted to the Department of Economics, Stockholm School of Economics (defended December 2024, awarded the highest grade) and awarded First Place in the 32nd Young Economist of the Year competition of the Czech Economic Society, 2025. The thesis as examined is in the school's library: https://hhs.primo.exlibrisgroup.com/permalink/46SSOE_INST/hejhp1/alma991001619398206056; Methods A to C and the appendix tables cited in the text are there. Section 3.1 adapts the export-status analysis of De Loecker and Warzynski (2012), and Section 3.2 applies the workflow of Imbens and Xu (2024) to these data; its design, estimator set and figures follow theirs. Of Section 3.3 only the last result is reproduced. I am grateful to my thesis supervisor, Jaakko Meriläinen, for invaluable guidance; to Matěj Bajgar, Mitch Downey, and David Schönholzer for their helpful feedback; and to Jiří Skuhrovec of Datlab s.r.o. for providing administrative data on government procurement. The thesis builds on my earlier work at the Institute of Economic Studies, Charles University Prague (https://dspace.cuni.cz/handle/20.500.11956/184831), where I accessed firm financial statement data via MagnusWeb. Replication files: https://github.com/marek-chadim/thesis.

1 Introduction

Public procurement constitutes approximately 12% of GDP in OECD countries, representing a substantial share of government expenditure (OECD, 2021). Ensuring its efficiency is vital for reducing resource waste and maximizing taxpayer value. Existing literature has examined key issues in public procurement, including discretion, political favoritism, and limited competition. However, much of this research has focused on comparing tenders, with relatively little attention devoted to differences between public procurement and private market outcomes.

This paper fills this gap by constructing a novel dataset and proposing methodological solutions to the identification challenges inherent in studying the dynamics of firm markups as firms enter and exit public procurement markets. To the best of my knowledge, this is the first study to provide credible econometric evidence on the relationship between markups and participation in public procurement. The findings contribute to ongoing debates on public procurement efficiency, market competition, and the implications of government contracts for firm-level market power.

Using the methodology of De Loecker and Warzynski (2012), I estimate markups under the assumption that cost-minimizing producers optimize variable input expenditures. This approach provides firm-level markup estimates without relying on assumptions about firms' competitive behavior in product markets. The results document the evolution of aggregate markups, showing a decline from 44% above marginal cost in 2006 to 31% in 2021, driven primarily by firms at the upper end of the markup distribution.

Next, I estimate the public procurement premium using non-experimental methods under the assumption of unconfoundedness, conditioning on information available prior to firms' engagement in public procurement. To mitigate biases arising from observational characteristics, I apply methodological advancements from Imbens and Xu (2024). Placebo tests, which compare the average markups of treated and control firms before the treated firms receive contracts, and sensitivity analyses benchmarking the omitted variable bias further reinforce the robustness of the findings.

Finally, I employ panel data models for binary interventions in longitudinal settings to estimate the causal effect of public procurement on the markups of government contractors, drawing on Arkhangelsky and Imbens (2024). These methods extend traditional difference-in-differences and two-way fixed-effects models by incorporating factor models, interactive fixed-effects models, and synthetic control approaches. The analysis emphasizes heterogeneity in causal effects and uses valid inference techniques suited to cases with a small number of treated units.

Drawing causal inferences from observational data, even with advanced methods, requires strong assumptions. The lack of natural experiments, instrumental variables, or regression discontinuity designs limits the availability of exogenous variation in this study. To address these challenges, I use a selection-on-observables approach, framed within the potential outcomes framework, to model counterfactual markups for government contractors in the absence of public procurement. The stable unit treatment value assumption (SUTVA) excludes interference between units and the dynamic effects of past treatments and outcomes. However, in competitive public procurement markets, violations of SUTVA can complicate the interpretation of causal effects. Despite this, the study prioritizes internal validity through a design-based balanced panel approach while enhancing generalizability by leveraging the full dataset. Balanced panel estimates, based on selection on observables, indicate that procurement raises contracting firms' markups by an average of 15%, with the full panel data model showing a decline in this effect from 30% in 2006 to 10% in 2021.

Industrial organization models with heterogeneous producers and firm specific markups provide potential explanations for these findings. More productive firms may absorb the costs associated with entering public procurement markets, which could explain why government contractors typically exhibit higher markups. Additionally, contractors may charge higher markups if they supply higher-quality goods produced using higher-quality inputs. Another plausible explanation is that procurement contracts often provide firms with long-term clients, leading to less elastic realized demand and enabling higher markups.

The literature on public procurement also highlights other mechanisms, such as discretion and political favoritism. My findings suggest that the public procurement markup premium has declined over time, consistent with reforms targeting single-bidding practices (Titl, 2023) and increased monitoring associated with European Union co-financing (Baránek and Titl, 2024).

The remainder of the paper is structured as follows. Section 1.1 reviews related work. Section 2 introduces the dataset and the construction of the public procurement indicator. Section 3.1 documents the distribution of markups and their descriptive relationship with public procurement. Section 3.2 presents the causal results under the unconfoundedness assumption, and Section 3.3 reports the full-panel estimates over time. Section 4 concludes. Methods A to C and the appendix tables referred to in the text are in the thesis.

1.1 Related Work

Public Procurement A central challenge in public procurement is navigating the trade-off between the benefits of discretion and the risks of rent-seeking. When the interests of procuring officials diverge from those of the public, discretion can heighten the risk of corruption and lead to the misallocation of contracts to politically connected firms. Conversely, auctions, while more transparent, are often more costly and time-consuming to organize compared to less formal procedures like direct negotiations. In the Czech Republic, Titl (2023) reports that approximately 23% of public contracts are awarded through single-bid procedures and that a 2012 reform aimed at reducing single-bidding led to a decrease in procurement project prices by about 6% of their estimated costs. Similarly, Kang and Miller (2022) find that single-bidding is prevalent in the United States but argue that information asymmetries and noncontractible quality dimensions can necessitate some level of discretion to incentivize effective contract execution, leaving the net impact of discretion on procurement outcomes ambiguous. In the Czech context, Palguta and Pertold (2017) document that officials frequently manipulate procurement thresholds by adjusting contract values to circumvent competitive bidding. However, identification challenges associated with bunching

complicate empirical assessments. Using a policy reform in Hungary, Szucs (2024) demonstrates that increasing discretion for public agencies results in higher prices, less productive contractors, and a greater prevalence of politically connected winners. Similar findings emerge in Italy, where Decarolis et al. (2020) show that discretionary procedures in government contracts, particularly those involving fewer bidders, heighten corruption risks and disproportionately benefit politically connected firms. In the Czech Republic, Baránek and Titl (2024) reveal that politically connected firms secure contracts priced 6% higher than competitively awarded contracts, with no corresponding improvement in quality. This misallocation aligns with evidence from Italy, where Bandiera et al. (2009) find that weaker governance leads public entities to overpay for comparable goods.

Markup Econometrics A markup can be estimated from the demand side or from the production side. The demand approach estimates demand elasticities, assumes a model of competition, and backs out marginal cost from the firms' pricing conditions. It needs prices and quantities by product, instruments for demand and a functional form for the demand system, and its answer depends on the conduct assumed. In return it supports counterfactual analysis, which the production approach does not. For measuring markups in retrospect the two can stand in for each other: in US brewing, where both have been applied, they give similar markups (De Loecker and Scott, 2016). De Loecker and Warzynski (2012) (DLW) introduced a method for estimating firm-level markups directly from production data. By combining input-output elasticities with input revenue shares, this approach captures market power in both product and factor markets. It overcomes many limitations of demand-based models by relying on more readily available production data. However, the DLW method has its own limitations. One critique is its assumption of Hicks-neutral productivity, where productivity shifts equally across all inputs. Raval (2023) shows that this assumption, combined with labor market frictions like hiring costs or monopsony power, can introduce bias by treating inputs such as labor and materials similarly. Bond et al. (2021) highlight that the DLW approach suffers from "omitted price bias" when using revenue data rather than actual output quantities, as revenue-based estimates fail to account for firm-specific price variations. Despite these criticisms, De Ridder et al. (2022) demonstrate that revenue-based markups still reveal im-

portant trends over time and across firms, making the DLW approach valuable for studying market power patterns, even if markup level estimates require careful interpretation. The production approach needs one assumption about behavior: in each period the firm chooses at least one variable input, free of adjustment costs, to minimize cost. Following DLW, write the production function as $Q_{it} = Q_{it}(X_{it}^1, \ldots, X_{it}^V, K_{it}, \omega_{it})$, where the X_{it}^v are variable inputs bought at prices P_{it}^v, K_{it} is capital and ω_{it} is productivity. Let λ_{it} be the multiplier on the output constraint in the cost minimization problem; it measures marginal cost. The first-order condition for input v is $P_{it}^v = \lambda_{it}\, \partial Q_{it}(\cdot)/\partial X_{it}^v$. Written as an elasticity, with the markup defined as $\mu_{it} \equiv P_{it}/\lambda_{it}$ for output price P_{it}, it reads

$$\frac{\partial Q_{it}(\cdot)}{\partial X_{it}^v}\frac{X_{it}^v}{Q_{it}} = \mu_{it}\frac{P_{it}^v X_{it}^v}{P_{it} Q_{it}}. \tag{1}$$

The left side is the output elasticity of input v, θ_{it}^v, and the fraction on the right is the input's share of revenue, α_{it}^v, so that $\mu_{it} = \theta_{it}^v/\alpha_{it}^v$. The share is read from the accounts. The elasticity is not observed and has to be estimated from a production function.[2]

2 Data

I utilize a dataset obtained through licensed[3] access to the MagnusWeb database[4], which provides full company accounts for an unbalanced panel of 1,297 firms active in the Czech construction sector. The dataset contains 7,261 observations covering the period from 2006 to 2021. The sample size accounts for the markup estimation requirement of having at least one lag of data. For robustness of the production function estimates, the top and bottom percentiles of firms are trimmed based on the sales-to-cost-of-goods-sold ratio.

In the Czech Republic, public procurement contracts are awarded under nationwide regulations requiring the publication of contract details by procurers. The dataset includes contracts awarded

[2]DLW estimate it with a control function, in which an input demand is inverted to stand in for unobserved productivity, as in Olley and Pakes (1996) and Levinsohn and Petrin (2003). I follow Ackerberg et al. (2015), whose timing assumptions on inputs identify all production function coefficients in the second stage; see Methods Section A.

[3]Charles University Prague, Faculty of Social Sciences, Institute of Economic Studies

[4]Dun & Bradstreet Czech Republic, a.s.

by central, regional, and municipal governments, as well as government-owned enterprises. A private company[5] corrected and cleaned the raw data to ensure its accuracy. I link firm financial statements with public procurement records to construct a binary indicator of government tender sales. This classification identifies firms that operate exclusively in the private sector, those that enter public procurement, those that exit it, and those that remain as government contractors.

Year	No. Firms	Share Contractors
2006	227	0.38
2007	290	0.33
2008	348	0.32
2009	412	0.32
2010	497	0.28
2011	506	0.26
2012	457	0.33
2013	235	0.41
2014	243	0.40
2015	245	0.48
2016	338	0.51
2017	660	0.57
2018	708	0.55
2019	764	0.53
2020	769	0.55
2021	562	0.53
Total	7,261	0.44

3 Results

Section 3.1 documents the main patterns of markups in the Czech construction sector and the relationship between markups and public procurement. I validate these results using two treatment effect identification strategies.

Section 3.2 employs a design-based approach relying on unconfoundedness, or selection based on observable financial statement information and pre-procurement markups. In contrast, Section 3.3 focuses on imputing the counterfactual markup evolution for government contracting firms, addressing selection on latent unobservable factors and incorporating synthetic control methods.

[5] Datlab s.r.o.

3.1 Markups and Public Procurement in the Czech Construction Sector

I use the empirical framework of De Loecker and Warzynski (2012)[6] to correlate markups with a firm's public procurement status and evaluate whether they change upon entering public procurement, controlling for input usage and aggregate trends. The regressions below adapt their analysis of exporters to public procurement. While I remain agnostic about a specific theoretical model, I draw on various economic mechanisms to interpret the findings. To estimate markups in the context of public procurement, I incorporate a firm's public procurement status into input demand equations, allowing it to directly affect the law of motion for productivity. After estimating the output elasticity of the variable inputs, I compute the implied markups using the first-order conditions leading to Equation 1.

The Evolution of Markups in Czech Construction Equation 1 defines the markup as the product of the output elasticity, θ, and the inverse of the variable input's revenue share, $\frac{PQ}{P^v X^v}$. The revenue share is directly observed in firms' income statements, while the output elasticities are estimated. These elasticities are both firm- and time-specific, which reflect technological differences across firms and changes over time. The average markup is calculated as: $\mu_t = \sum_i m_{it}\mu_{it}$, where m_{it} represents the weight of each firm. I use the share of sales as the weight and compare it with total costs as the input weight. Figure 1 shows the evolution of average sales-weighted and input-weighted aggregate markups in the Czech construction sector. Early in the sample period, markups were stable at around 1.45, declined to approximately 1.25 during 2012–2014, and rose slightly to just above 1.35 by the early 2020s. In 2006, the average markup stood at 44% above marginal cost, compared to 31% in 2021. However, averages alone fail to capture the changes in the markup distribution. The strength of the De Loecker and Warzynski (2012) method lies in its ability to estimate firm-specific markups, which enables an analysis of the entire distribution. A key finding is that a small subset of firms drive the overall decline in markups, while most firms see only modest decreases.

[6] See Methods A for markup and production function estimation details.

Figure 1: Average Markups. Estimated output elasticities $\hat{\theta}_{it}^{V}$ are time- and firm-specific.

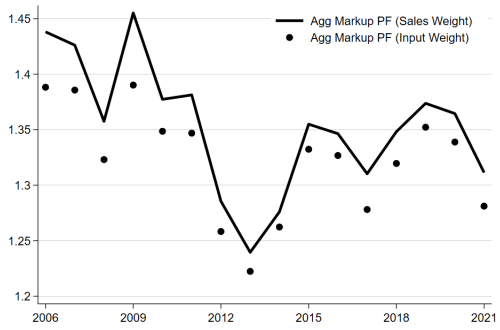
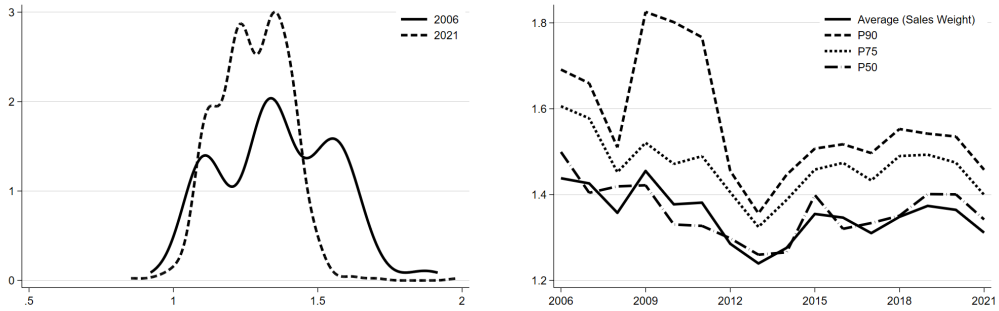


Figure 2: The Distribution of Markups $\hat{\mu}_{it}$



(a) Kernel Density (unweighted)

(b) Percentiles (revenue-weighted)

Figure 2a illustrates the shifts in the distribution by showing the kernel density of the unweighted markups for 2006 and 2021. Over time, the variance of markups has narrowed, driven primarily by a thinning and shortening of the upper tail. Appendix Table 5 provides the unweighted markup distribution by year.

Because the kernel density does not account for firm weights, Figure 2b shows the moments of the sales-weighted markup distribution over time. Firms are ranked by markup, and percentiles are weighted by each firm's market share. This weighting ensures comparability between the percentiles and the share-weighted average. The rankings are updated annually, meaning that the top firms may vary from year to year. Firms in the upper half of the markup distribution primarily drive the decline in average markups. The median (P50) and lower percentiles remain mostly stable over time. The sharpest drop occurs at the 90th percentile, particularly before 2012, when markups fell significantly from 1.8 to 1.4. This indicates that the overall reduction in average markups is due to a few firms experiencing substantial declines compared to earlier periods.

This finding provides preliminary evidence of the evolving relationship between markups and public procurement. It is particularly relevant when viewed alongside Titl (2023), who demonstrated that the 2012 Czech public procurement reform, which eliminated single-bid contracts, reduced prices relative to estimated costs for these contracts.

Do Government Contractors Have Different Markups? I examine whether government contractors systematically have different markups compared to private-sector firms by using firm-specific markups in a regression framework. Specifically, I estimate the percentage difference in markups between government contractors and private-sector firms.

This approach ensures robustness even if the variable inputs used to compute markups are subject to adjustment costs, provided that such costs do not disproportionately affect government contractors.

The regression specification is as follows:

$$\ln \mu_{it} = \delta_0 + \delta_1 \, pp_{it} + \mathbf{b}_{it}'\sigma + \nu_{it}, \tag{2}$$

where pp_{it} is a dummy variable indicating a firm's public procurement status, and δ_1 measures the percentage markup premium for government contractors. The vector $\mathbf{b}_{it}$ includes control variables[7], with σ as their corresponding coefficients.

It is important to note that δ_1 is not interpreted as a causal parameter. Instead, this specification identifies whether government contractors, on average, have different markups. To the best of my knowledge, this relationship has not previously been documented, making these findings a novel contribution. While the coefficients of the control variables are not the primary focus of this analysis, I revisit the correlation between markups and other economic factors later in the paper. The regression is estimated on the full sample of construction firms with full year–subindustry interaction dummies. The percentage premium parameter estimate, $\hat{\delta}_1$, is 0.149, with a standard error of 0.003.

[7]Control variables include variable input and capital use to account for differences in size and factor intensity, along with 47 year–subindustry interaction dummies to capture aggregate markup trends.

These results suggest that government contractors, on average, charge 14.9% higher markups than private-sector firms. They align with microeconomic models in which government contractors charge higher markups due to greater productivity, allowing them to outcompete rivals in tender processes. A comparison of the average markups between government contractors and private-sector firms supports this prediction and highlights its potential policy implications. Models with heterogeneous firms suggest that market share reallocates from less efficient producers to more efficient ones, with government contractors expected to exhibit higher productivity. This higher productivity enables them to cover the fixed costs associated with entering public procurement markets. However, because measured productivity is a residual of a sales-generating production function, it likely also captures unobserved differences between government contractors and private-sector firms. As such, caution is warranted when interpreting the public procurement-productivity relationship.

Public Procurement Entry and Markup Dynamics My dataset allows me to examine how markups evolve for firms entering public procurement markets, both before and after becoming government contractors. I focus on three distinct groups of government contractors identified in the sample: starters, quitters, and continuous contractors[8]. To analyze these groups, I estimate the following regression to compare markups before and after public procurement entry and exit and to measure the markup differential for firms that consistently receive government contracts:[9]

$$\ln \mu_{it} = \gamma_0 + \gamma_1 \text{Entry}_{it} + \gamma_2 \text{Exit}_{it} + \gamma_3 \text{Always}_i + \mathbf{b}'_{it}\sigma + \nu_{it}, \tag{3}$$

where $\text{Entry}_{it} = 1$ if a firm becomes a government contractor in year t, and $\text{Exit}_{it} = 1$ if a firm stops contracting with the government. The constant term, γ_0, represents the average log markup for private-sector firms, including those that enter or exit public procurement markets. The primary coefficients of interest are γ_1, which captures the percentage change in markups for starters (firms

[8] $\text{Entry}_{it} = \mathbb{1}\{\Delta pp_{it} = 1\}$, $\text{Exit}_{it} = \mathbb{1}\{\Delta pp_{it} = -1\}$, $\text{Always}_i = \mathbb{1}\{\frac{1}{16}\sum_{t=1}^{16} pp_{it} = 1\}$

[9] I exclude 152 firms that enter and exit public procurement markets more than once during the sample period to ensure cleaner comparisons.

entering public procurement) between the pre- and post-entry periods; γ_2, which measures the impact of public procurement exit on markups for quitters; and γ_3, which reflects the markup premium for continuous contractors, firms consistently engaged in public procurement throughout the sample period.

My results indicate that public procurement entry is associated with a significant markup increase of approximately 12%, while controlling for aggregate markup dynamics with year and subindustry dummies. All coefficients are reported in Appendix Table 8.

These results suggest that firms entering public procurement experience substantial markup increases, highlighting the role of public contracts in shaping firm behavior. The markup premium for continuous contractors, captured by γ_3, further suggests that sustained public procurement engagement allows firms to maintain higher pricing power over time.

Discussion This section reports two main findings: (i) in the cross-section, government contractors exhibit higher markups than their private-sector counterparts within the same industry, and (ii) in the time series, markups increase when firms enter public procurement markets, even after controlling for costs of goods sold, capital, and aggregate demand and supply effects using subindustry-year dummies. Government contracting firms tend to display higher measured total factor productivity, as reflected in the gap between observed revenues and predictions from estimated production functions. More efficient producers may charge lower prices, capture larger market shares in private markets, and outperform rivals in government tenders. Their cost advantage allows them to set higher markups under certain conditions, especially when their relative efficiency translates into competitive advantages in both private and public procurement markets. Alternatively, public procurement markets may differ from private markets in terms of demand elasticities or government valuations of goods and services.

However, the exact mechanism driving these results cannot be conclusively identified with the available data, as I lack firm-specific price information to disentangle markup differences arising from cost versus price effects. Nonetheless, the observed within-firm increase in markups for Czech government contractors suggests that factors beyond cost differences influence pricing behavior.

This aligns with findings by Titl (2023), Baránek and Titl (2024), and Szucs (2024), who emphasize the roles of discretion and favoritism in public procurement.

In summary, firms supplying construction projects to the Czech government charge higher markups than firms operating solely in the private sector, and their markups increase significantly upon entering the public procurement market.

Robustness to Omitted Prices In estimating output elasticities,[10] I use deflated sales as a proxy for physical quantities, which may introduce omitted variable bias from unobserved prices, as noted by Klette and Griliches (1996). For price-taking firms, this is not an issue, as Equation 1 can be rewritten in terms of sales: $\partial R_{it}(\cdot)/\partial X_{it}^v = P_t \cdot \partial Q_{it}(\cdot)/\partial X_{it}^v$. However, for firms with market power, unobserved prices can bias output elasticity estimates: $\partial R_{it}(\cdot)/\partial X_{it}^v = \partial Q_{it}(\cdot)/\partial X_{it}^v \cdot (P_{it} + \partial P_{it}/\partial Q_{it})$. Such bias may arise if increased input usage reduces input prices under standard demand and cost specifications, leading to downward bias in elasticities and underestimated markups. This primarily affects the levels of estimated elasticities rather than the relationship between markups and public procurement status. The average percentage markup difference is consistent as long as deviations between estimated and true elasticities are not systematically correlated with public procurement status. To mitigate this issue, I estimate markups using a productivity proxy, $h(\cdot)$, which accounts for price variation correlated with productivity. While unrelated demand shocks may still bias input coefficients, they do not affect the core relationship. In regressions, I use the dependent variable $\ln \hat{\theta}_{it}^V - \ln \hat{\alpha}_{it}^V$, where $\hat{\alpha}_{it}^V$ is the expenditure share, and price bias affects output elasticities $\hat{\theta}_{it}^V$. I employ a translog production function: $\hat{\theta}_{it}^V = \theta_{it}^V + \rho(v_{it}, k_{it})$, where v_{it} and k_{it} are the logarithms of expenditures on variable inputs and capital, and $\rho(\cdot)$ captures unobserved price deviations. I control for inputs v_{it} and k_{it} in both Equations 2 and 3, as in Bond et al. (2021, Appendix B.2).

[10] See Methods A.

3.2 Estimation of the Public Procurement Effect Under Unconfoundedness

This section estimates the effect of holding a public contract on a firm's markup in the contract year, under the assumption that selection into contracts depends only on what I observe. The design follows the workflow that Imbens and Xu (2024) recommend for observational studies: estimate the propensity score flexibly, check overlap and trim, compare several estimators including doubly robust ones, and test the identifying assumption with placebo outcomes and a sensitivity analysis. The two assumptions behind it, unconfoundedness and overlap, are stated in Methods B.

I do not observe how contracts are assigned. What I do observe is each firm's markup history, which I use both to adjust for selection and to test the adjustment. To have a complete history for every firm, I restrict the sample to a balanced panel as a design-based decision: firms observed in all 16 years, excluding those that already held public contracts in 2006. This leaves 26 firms. The covariates need four lags, so the estimation sample covers 2010–2021 and has 312 firm-year observations.

The treatment is binary: $W_{it} = 1$ if firm i has revenue from public tenders in year t. There are 151 treated firm-years. The 161 control firm-years are those without such revenue: 60 belong to five firms that hold no public contract in 2010–2021, and 101 are years in which the other 21 firms hold none. The 21 firms differ in how often they contract: two do so in a single year, three in all twelve, and the median firm in seven. The contracts in the sample total 25.9 billion Czech crowns over 2010–2021, about 1.2 billion USD at the 2021 exchange rate (1 USD = 21.5 CZK).

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total
Contract Value (Billions CZK)	2.20	0.27	0.60	2.82	1.11	0.84	0.59	1.99	5.05	3.84	2.98	3.60	25.90
Number of Contracting Firms	7	8	9	13	11	14	12	17	17	15	15	13	151

The outcome is the log markup in the contract year, $Y_{i,0}$; I also report the two following years, $Y_{i,1}$ and $Y_{i,2}$. The windows for testing and for adjustment are those of Imbens and Xu's lottery application. The log markups of the three years before the contract, $Y_{i,-3}, Y_{i,-2}, Y_{i,-1}$, and their average are placebo outcomes: a contract cannot affect them, so an estimated effect on them would signal selection that the adjustment has missed. The log markups of the three years before that,

$Y_{i,-6}, Y_{i,-5}, Y_{i,-4}$, are covariates, together with the firm's sales, cost of goods sold, assets and number of employees four years earlier, its public procurement status in each of the four preceding years, the calendar year and the NACE 2-digit division. Where a lagged or later markup falls outside the panel, the code sets it to zero; this affects the first two years of the covariates and the last two years of the later outcomes.

Table 1 compares treated and control firm-years on covariates measured four years earlier. Contract years belong to larger firms with higher markups: the difference in lagged log markups is 0.64 pooled standard deviations, and the difference in employment is 0.99. Differences above 0.25 are generally considered substantial (Imbens and Wooldridge, 2009), and with gaps of this size a linear regression leans on extrapolation and its result depends on functional form. Each estimator below uses the control firm-years to predict what a treated firm-year's markup would have been without the contract, $\mu_0(x) \equiv E[Y_i(0) \mid X_i = x]$.

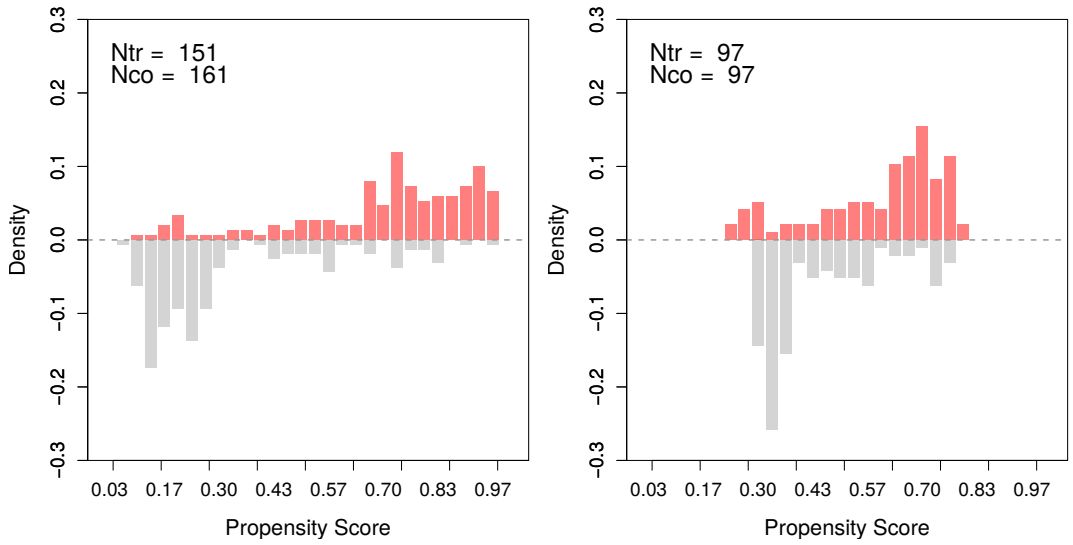
Table 1: Pretreatment Summary Statistics

Observations Stratified by Engagement in Public Procurement	$\{i,t;W_{it}=0\}$ N=161 **mean (s. d.)**	$\{i,t;W_{it}=1\}$ N=151 **mean (s. d.)**	**diff / sd**
No. Employees	52.57 (42.03)	104.96 (61.67)	0.993
$\ln(\text{Sales})_{t-4}$	18.52 (1.01)	19.27 (0.85)	0.799
$\ln(\text{Costs})_{t-4}$	17.89 (0.95)	18.52 (0.82)	0.713
$\ln(\text{Markup})_{t-4}$	0.24 (0.13)	0.34 (0.17)	0.640
$\ln(\text{Assets})_{t-4}$	16.05 (1.68)	16.97 (1.37)	0.603
NACE 42	0.04 (0.20)	0.11 (0.32)	0.259
NACE 43	0.40 (0.49)	0.44 (0.50)	0.081

Note: Firm-years from the balanced panel of 26 firms; 13 firms already active in public procurement in 2006 are excluded. The years 2006 to 2009 supply the lags. $N \times T = 312$ observations in 2010–2021. The measure diff/sd is the absolute difference in means divided by the pooled standard deviation:
diff/sd $= \left|\bar{X}_0 - \bar{X}_1\right| \left(\sqrt{(s_0^2 + s_1^2)/2}\right)^{-1}$.

Figure 3 plots the estimated propensity scores, from a generalized random forest, for treated and control firm-years. Most treated observations have scores that control observations also reach, but treated observations with scores above 0.8 have almost no counterparts.

Figure 3: Propensity Scores Before and After Trimming



(a) Full sample (b) Trimmed sample

Note: Propensity scores estimated with a generalized random forest and re-estimated after trimming in (b). Treated firm-years above the axis, control firm-years below. Figure design after Imbens and Xu (2024).

Following their trimming procedure, I drop observations with a score above 0.8, which removes 53 treated and 8 control firm-years, and then match each remaining treated observation to one control observation on the re-estimated score, without replacement. The matched sample has 97 treated and 97 control firm-years.

Table 2 reports the average effect on the treated in the matched sample. Its rows are the eleven estimators that Imbens and Xu (2024) compare: the raw difference in means; three outcome models (linear regression, Oaxaca–Blinder, and a generalized random forest, GRF); two matching estimators (five nearest neighbors with bias correction, and the propensity score); three weighting estimators (inverse probability weights from GRF scores, covariate balancing propensity scores, and entropy balancing); and two doubly robust estimators (double machine learning with an elastic net, and augmented inverse probability weighting with GRF, AIPW-GRF). Across the eleven rows the contract-year estimate lies between 0.12 and 0.16, about 15% on average, in line with the descriptive premium of Section 3.1: holding a public contract raises a firm's markup in the contract year.

The last two rows are the ones to read for the placebo test: their estimates for the pre-contract markup are -0.01 and 0.00. The standard errors are heteroskedasticity-robust and treat firm-years as independent; with 26 firms, errors clustered by firm would be wider.

Table 2: Contract-Year and Placebo Estimates, Matched Sample

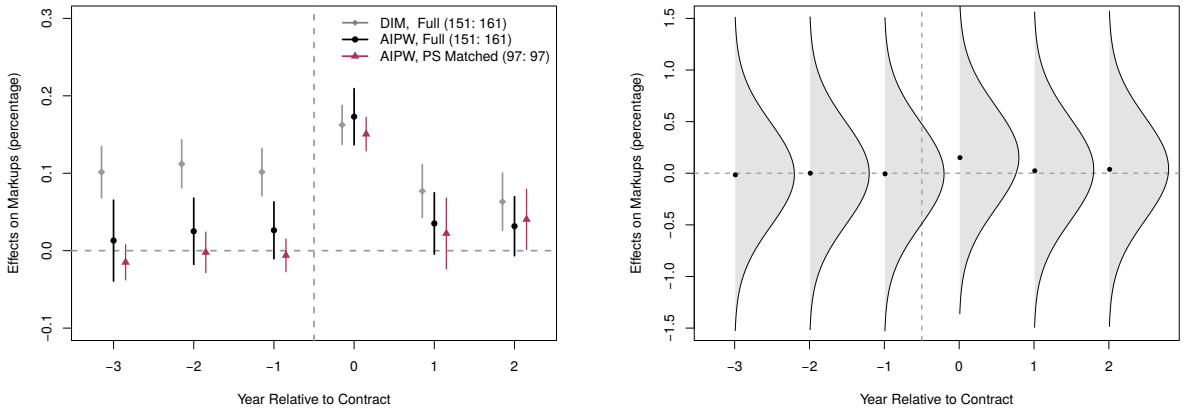
Effect on Markups	**Contract**	**Pre-Contract Average**
Difference-in-Means	0.12 (0.02)	0.03 (0.02)
Regression	0.16 (0.01)	-0.00 (0.01)
Oaxaca Blinder	0.15 (0.01)	0.00 (0.02)
GRF	0.13 (0.01)	0.03 (0.01)
NN Matching	0.15 (0.01)	0.01 (0.01)
PS Matching	0.13 (0.01)	-0.00 (0.01)
IPW	0.14 (0.02)	0.01 (0.02)
CBPS	0.15 (0.02)	0.00 (0.02)
Entropy Balancing	0.15 (0.03)	-0.00 (0.02)
DML-ElasticNet	0.16 (0.01)	-0.01 (0.01)
AIPW-GRF	0.15 (0.01)	0.00 (0.01)

Note: ATT estimates in log points on the trimmed and matched sample (propensity scores above 0.8 excluded). First column: log markup in the contract year. Second column: the placebo outcome, the average log markup of the three years before the contract. Robust standard errors in parentheses. Estimator set after Imbens and Xu (2024).

Figure 4 shows the same comparison for each year from three before the contract to two after. Before the contract the raw difference in means is already about 0.10, the selection that Table 1 shows, while the AIPW-GRF estimates are close to zero in all three pre-contract years, in the full sample and in the matched one. The plot follows Imbens and Xu's figure for the lottery data and, as they note of theirs, it is not an event study: each point is a separate cross-sectional comparison under unconfoundedness, and no parallel-trends assumption is involved.

An average can hide differences along the markup distribution. Figure 5 therefore reports effects on the treated quantile by quantile (QTET), from the weighting estimator of Firpo (2007), for the placebo outcome and for the contract-year markup, in the full sample and in the matched sample. In the matched sample the adjusted placebo effects cannot be told from zero at any quantile, and the contract-year effects are positive across the distribution. In the full sample the adjusted placebo effects are positive above the median, which is the selection that trimming removes.

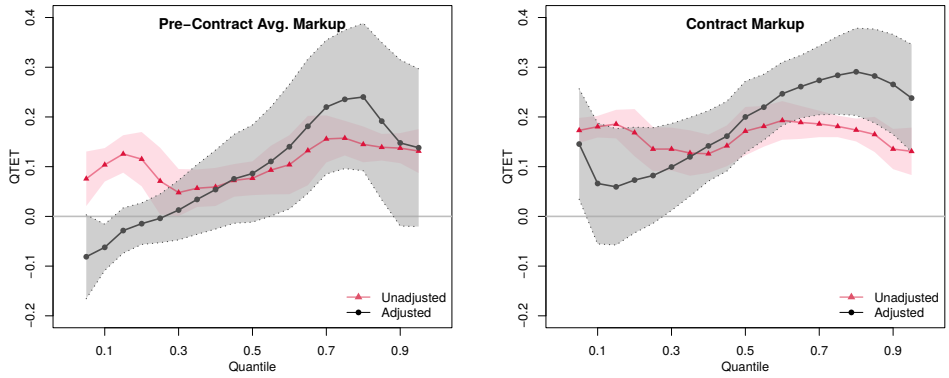
Figure 4: Estimates by Year Relative to the Contract



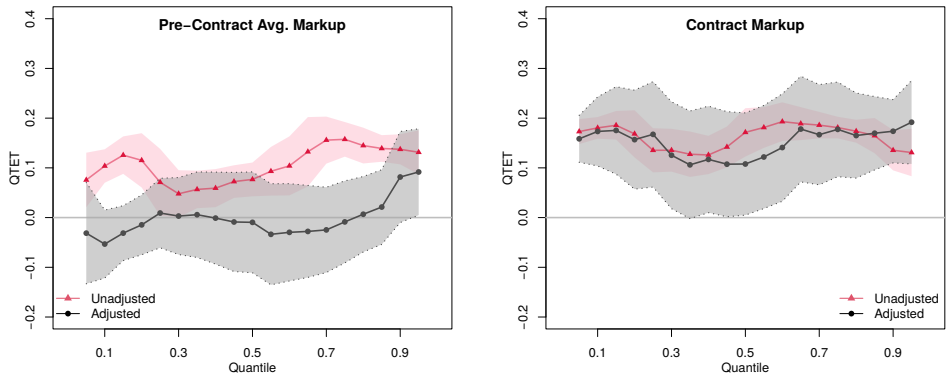
(a) ATT (b) CATT

Note: Panel (a): ATT in log points by year relative to the contract, with 95% confidence intervals. Gray: difference in means, full sample. Black: AIPW-GRF, full sample. Red: AIPW-GRF, matched sample. Years –3 to –1 are placebo outcomes. Panel (b): spread of the conditional effects (CATT) from AIPW-GRF in the matched sample, by year; the dots are the ATT. Figure design after Imbens and Xu (2024).

Figure 5: Quantile Effects on the Treated



(a) Contractors vs Controls (Full Sample)

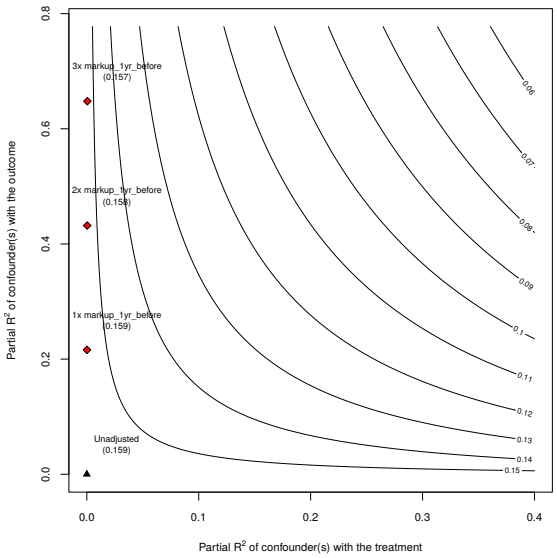


(b) Contractors vs Controls (Trimmed Sample)

Note: QTET by quantile with bootstrapped 95% confidence bands. Pink: no covariates. Gray: weighted by a logit propensity score on all covariates. Top row: full sample. Bottom row: matched sample. Left: the placebo outcome. Right: the contract-year log markup. Figure design after Imbens and Xu (2024).

The estimate is also stable when I allow for an unobserved confounder. In the sensitivity analysis of Imbens (2003), as extended by Cinelli and Hazlett (2020), the regression estimate with all covariates is 0.159, and a confounder three times as strongly related to treatment and to the outcome as the markup of the year before the contract would leave it at 0.157 (Figure 6).

Figure 6: Sensitivity to an Unobserved Confounder



Note: Contours give the regression estimate of the effect under a hypothetical confounder with the partial R^2 on the axes (Cinelli and Hazlett, 2020). Diamonds mark confounders one, two and three times as strong as the log markup one year before the contract; the triangle is the estimate without a confounder. Plot from the sensitivity tools of Cinelli and Hazlett (2020).

Taken together, the pre-contract estimates give no sign of remaining selection in this sample. That is as far as the test goes: it speaks to selection on past markups, not to anything that changes in the contract year itself.

3.3 Model-based Estimation of the Public Procurement Effect

This section of the thesis extends the analysis from the balanced panel to the full sample and to selection on unobserved factors. It imputes each contractor's markup without the contract from fixed-effects, interactive fixed-effects and matrix completion models (Liu et al., 2024; Athey et al., 2021). On the balanced panel of Section 3.2, the fixed-effects estimate of the contract-year effect is 0.20. Only the last result of the section is reproduced here.

I present the year-aggregated ATT estimates from the matrix completion method in Table 3. The results provide further evidence of a decreasing trend in the public procurement premium over time for firms in the Czech construction sector. Specifically, my estimates suggest that, compared to firms operating only in the private sector, government contractors had markups approximately 30% higher from 2006 to 2009, over 20% higher during the period 2010–2015, and about 10% higher during 2016–2021, only one-third of their initial level.

Table 3: Matrix Completion Year-Aggregated ATT Estimates

Year	ATT	Standard Err.	No. Treated
2006	0.294	0.016	12
2007	0.282	0.022	7
2008	0.300	0.014	8
2009	0.337	0.010	18
2010	0.273	0.017	9
2011	0.241	0.012	11
2012	0.236	0.013	18
2013	0.262	0.010	18
2014	0.301	0.008	20
2015	0.219	0.008	31
2016	0.092	0.008	34
2017	0.105	0.008	77
2018	0.100	0.006	68
2019	0.101	0.006	74
2020	0.118	0.007	78
2021	0.098	0.008	58

4 Conclusion

Using firm-level data on firms in the Czech construction sector, this study employs a method that accommodates diverse price-setting strategies and nonconstant returns to scale to estimate firm-specific markups. It describes the evolution of market power through the distributional characteristics of markups. Between 2006 and 2021, markups declined from 44% to 31% above marginal cost, a decline primarily driven by firms with the highest initial markups. Over time, the distribution of markups became less skewed, with a thinner upper tail.

The decline in Czech construction aggregate markups is linked to the dynamics of the public procurement markup premium. The relationship is validated using policy evaluation designs to account for both observable and unobservable selection bias.

In the balanced sample, the year-of-contract-treatment effect is estimated to be 15% under selection on observable lagged variables, while the same-sample fixed-effects analysis yields a point estimate of 20%. Using both the unconfoundedness approach and the TWFE model to test robustness to alternative identifying assumptions leads to the bracketing property bounds on the estimated causal effect (Angrist and Pischke, 2009, Section 5.3).

The analysis also integrates Synthetic Difference-in-Differences re-weighting and matching on pre-exposure trends to reduce reliance on parallel-trend assumptions. This approach accounts for additive unit-level shifts while utilizing data-driven synthetic control methods.

Finally, the full sample estimation employs interactive fixed effects and matrix completion methods. The framework is validated using diagnostic tests of out-of-sample predictions for untreated potential outcomes, which are robust to model misspecification and overfitting. Subgroup analysis indicates that the overall effect is primarily driven by government contracts awarded in the early years of the sample. The estimated average public procurement effect on government contractors is 30% in 2006 and decreases to 10% by 2021.

These findings contribute to the public procurement literature by establishing a relationship between markups and public procurement. They also provide a benchmark for assessing the competitiveness of government projects relative to those in the private sector. The results provide empirical support for reforms aimed at eliminating single-bidding practices and increasing transparency in the tendering process that prevents favoritism in public procurement. They offer policy-relevant insights, highlighting the role of institutional improvements in enhancing the efficiency of public spending.

Limitations and Future Directions This study has several limitations related to its methodology and the available data, highlighting opportunities for refinement and further research.

A key limitation is the reliance on deflated sales as a proxy for physical quantities in estimating output elasticities. While practical given the data constraints, this approach may introduce omitted variable bias due to unobserved price variations, potentially distorting output elasticity and markup estimates. Access to firm-specific price data or product-level sales would improve the precision of these estimates by disentangling the effects of costs and pricing strategies. Such data would also enable a more detailed analysis of how public procurement influences firm-level pricing behavior.

The lack of exogenous variation in the data such as from natural experiments or valid instrumental variables presents another limitation, restricting the ability to fully address the endogeneity of firms' engagement in public procurement. Although this study employs selection on observables and unobservables to mitigate these concerns, the approach remains vulnerable to selection bias. Furthermore, the stable unit treatment value assumption (SUTVA) may not hold in dynamic competitive procurement settings, where interactions between firms or cumulative effects of past participation can influence both current and future participation as well as associated outcomes.

Access to richer and more granular data would support alternative and more detailed analysis. For example, operational data such as hours worked, employee allocations across projects, or firm-specific cost structures could provide a clearer understanding of how firms adjust operations and pricing strategies across public and private markets. Additionally, contract-level details such as tendering procedures, competition intensity, and procurement award criteria would shed light on the institutional factors shaping markup differences. These data enhancements would also enable researchers to disentangle market-specific effects and better identify causal mechanisms.

Despite these limitations, the framework employed in this study offers a foundation for analyzing the relationship between markups and public procurement participation. Addressing the data and methodological constraints outlined here would enable future research to look more closely at the mechanisms driving markup differences. Such advancements would strengthen the evidence base for policy reforms aimed at improving the efficiency and transparency of government procurement processes.

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